CubeSat materials

November 3, 2026

Table 1: Bus materials. Contribution groups are colored by status: Flight-linked , Changed or superseded , Unvalidated , Ground reference , Not found .

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Command and data handling	EAGLE schematics and PCB, C3 XLSX BOM and historical CAN/backplane design. [1, 2] *C3 v5 is listed in OreSat0's 2022 flight manifest.* C3 v7.1 KiCad source, Linux C3 services, CANopen management and tests. [3, 4] *C3 v5 uses STM32/ChibiOS. Later software uses Linux/CANopen. The current v7.1 board links to a BOM labeled v6.1r0.* Preserved ChibiOS F4 application/control firmware tag c3_v1_1. [5]	KiCad schematic/PCB, obc.csv BOM, STM32/FreeRTOS code, ECSS routing/storage/time/scheduling services and test programs. [6–8] *OBC firmware tags upsat-fm-1.0 and upsat-fm-1.1 point to the same commit.*	Production-labeled FreeRTOS task, storage, sensor and command C source. [9, 10] *The microSD failure branch leaves alternative-memory storage as a TODO.* Altium schematics, Gerbers, schematic PDF, XLSX BOM and soldering checks labeled engineering model V2.1. [9]	Early mainboard schematic/PCB/BOM references and KickSat-2/V-R3x platform publications. [11–13] KiCad source, CSV/interactive BOMs, Gerbers, drivers and bootloader/firmware configuration. [14, 15] *v05c is dated June 2021, after V-R3x's January 2021 launch. The maintainer's preliminary 2024 tests report SAMD51 peripheral failures under 50 MeV protons. [13, 16]* Task, storage/logging, command and power-management example source/tests. [17] *Advanced command examples use fixed passcodes and remote eval/exec.*	OBC/ADCS KiCad reference generations, manufacturing files, C++/FreeRTOS, PUS services and unit tests. The README documents an internal watchdog disabled for debugging. [18, 19] *The 2026 paper reports OBC/ADCS EQM qualification after four environmental campaigns. [20]* Newer OBC/ADCS KiCad board release. [21] *The README labels v3.1 the flight model awaiting manufacture.*	OBC/EPS schematic PDFs, XLSX BOMs and PIC main/COM/reset command/storage/reservation firmware. [22, 23] *The platform ICD specifies fixed flight software during orbital operations. [24]* Separate firmware, CPLD and hardware/parts-list generation. [25] *The March 2023 review records zero successful uplinks, downlinks or received signals for all three BIRDS-5 spacecraft. [26]*
Communications	C3 v5 RF schematic/PCB/BOM and antenna/deployment reference. [1, 2] *An oscillating L-band LNA jammed reception. UHF firmware disabled its LNA, reducing receive sensitivity by about 35 dB.* v7.1 KiCad RF sheets and current radio/packet service source. [3, 4] *Radio silicon changed from AX5043 to AT86RF215.* DxWiFi and handheld image-reception hardware/software references. [2, 27]	Published v3/v3.1 hardware tags, CC1120 radio/antenna design and flight-model AX.25/CW firmware. [28–30] Inspected master KiCad source, Gerbers and antenna-release files. [31] *The master snapshot differs from the v3/v3.1 hardware release tags.*	Schematic PDF, BOM, beacon/radio firmware, antenna interface and technical manual. [9, 10, 32] *Flight decoding exposed AX.25 interoperability and Reed-Solomon issues. [33]* Additional Altium PCB source and Gerbers in the migrated TTC repository. [34] *The migrated schematic PDFs are byte-identical to the older release.*	Early RF/mainboard source and publications describing actual mission links. [11–13] *V-R3x demonstrated 253 kbps S-band and 37.5 kbps UHF crosslinks.* RF_and_GPS.sch, routed PCB/BOM and RFM9x driver source. [14, 15] *The v05 radio integration uses a commercial HopeRF RFM98PW module.* Test-monopole documentation, incomplete flight-antenna guide and Beep-Sat beacon/ACK examples. [17, 35]	SatNOGS-derived KiCad schematic/PCB, Gerbers, BOM, MCU/FPGA source, coding/modulation libraries and tests. [36, 37] *The MCU/FPGA task sends fixed test bytes. The 2026 paper reports COMMS EQM environmental qualification. [20]*	ADDNICS/Radiometrix ICDs, COM firmware, antenna-panel EAGLE/BOM and assembly material. [22, 23, 38] *The RF design uses commercial ADDNICS and Radiometrix modules.*

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Attitude determination and control	OreSat0 ACS/star tracker hardware references and mission card manifest. OreSat0.5 ADCS demonstration description. [2, 39, 40]	gEDA schematics/PCB, Main.bom, sensor and magnetic/spin-actuator interfaces, and actuator mechanical CAD. [41, 42] *Main.bom contains unknown part fields. Sun sensing uses an external NewSpace Systems module. [43]*	N45 magnet/hysteresis-bar descriptions and qualification/mounting lessons. [10, 45] *Passive stabilization uses one N45 magnet and four Permanorm 5000 H2 hysteresis bars.*	Published gyroscope results and the mission IMU description. [13] *V-R3x measured tumble rates above 30 degrees/s.*	Sensor interfaces, drivers, MEKF/B-dot/Sun/nadir algorithms, allocation and tests. Commercial wheel/iMTQ interfaces. [18, 46] *The shared OBC/ADCS EQM completed environmental qualification. [20]*	Mission publication reporting manual reaction wheel response and magnetorquer power-on. [23] *The estimator returned all-zero attitude data.*
	ADCS schematics/PCB files, some BOMs, older wheel CAD, drivers, estimation/control algorithms, simulations and tests. [4, 39] *C3 wheel-command writes remain TODOs. Both v8 wheel repositories contain README files only.*	Wahba estimation, B-dot/pointing control, actuation firmware and sensor-fusion test code. [43, 44]	MPU9250 schematic/interface/task source and older magnetorquer-driver artifacts. [9]	Avionics.sch/BOM with BMX160 and its driver. [14, 15] *The sensor changed from LSM9DS1 to BMX160.*	OBC/ADCS v3.1 hardware source. [21] *Board generations advanced from v2.5/v3 to v3.1.*	ICD passive-stabilization options and OBC sensor/mission-data handling. [22, 24]
				Beep-Sat IMU logging task and platform integration guidance. [17, 35]		
Guidance, navigation and control	Flight manifest and historical GPS hardware design describing Venus0838 and the MAX2771 experiment. [2, 47]	ADCS receiver interfaces and the thesis description of the PQNAV-L1 integration. [41, 43]	MPU9250 accelerometer schematic, imu.c interface and acquisition task source. [9]	Published mission GPS configuration and navigation experiment descriptions. [13] *GPS receivers used orbital firmware permissions. Complete ranging and radiation datasets remained onboard after contact ended.*	OrbitalParameters.cpp, SGP4/geodetic conversion, tests and AOCS design manual. [46, 49] *SGP4 propagates uploaded TLEs for attitude references.*	ICD ground-position/optional-GPS descriptions and mission firmware GPS-byte handling. [22–24] *BIRDS-4 GPS and SGP4 modes remained inactive.*
	Orion B16 parsing, CANopen position/velocity/time service, GPS card KiCad source and tests. [47, 48] *The receiver/software target changed from Venus0838 to Orion B16.*	NMEA/UTC parsing, TLE handling and SGP4 position/velocity propagation. [43, 44] *SGP4 output supplies the attitude reference model.*		GPS footprint/interface in the native mainboard schematic. [14, 15] *The v05 schematic marks GPS as DNI. Default GPS initialization is commented out.*		
	MAX2771-related hardware and a prototype GPS front-end CSV BOM. [47, 48] *The software README labels the SDR path unimplemented.*			GPS driver and NMEA example in the technical manual. [35]		
Structures and mechanisms	Native SolidWorks assemblies/parts, STEP exports, frame drawings/BOMs, card clamps and antenna restraint/deployment geometry. [2, 50]	SolidWorks assemblies/parts, STEP, aluminium/CFRP frame, battery mounts, antenna/spin-torquer geometry and deployment electronics/code. [31, 42, 51]	Board-stack diagrams, commercial ISIS chassis/antenna integration, mounting/test lessons and release C code. [9, 10, 45] *EM-II vibration exposed intermittent battery contacts. Flight assembly added connector reinforcement.*	V-R3x publication describing the modified commercial structure and deployment approach. [13] *The mission paper proposes unconstrained tape-antenna deployment as the cause of tumbling.*	Historical ADM mechanical STEP/STL, electronics, test hardware and failed-campaign report. [20, 53] *The in-house ADM was abandoned and the platform redesigned.*	BIRDS-3 SolidWorks assemblies/parts/drawings, BIRDS-4 STEP and platform battery/frame/restraint/assembly documentation. [24, 56, 57]
	Public Onshape design link replacing the deprecated SolidWorks repository. [50]			Board STEP, dimension drawing, PCB outline and burn-wire circuitry. Battery-board STEP also exists. [14, 52] *These files describe mainboard v05 and battery board v01b.*	Some native frame parts and simplified STEP geometry in thermal/design repositories. [49, 54] *The main public assembly repository contains a GitLab template README. [55]*	BIRDS-5 drawings and assembly/procedure material. [58, 59] *The CAD release provides PDF drawings and a screw list. Its README defers native Fusion files to a future cloud link.*
				Published burn-wire software and chassis mounting instructions. [15, 35]		

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Thermal	Conductive frame/contact design and Thermal Desktop/SINDA models/results for OreSat0 and OreSat0.5. [50, 60]	EPS heater/temperature schematic, battery CAD and thesis TVAC/integration descriptions. [42, 43, 63]	Battery heater construction descriptions, thermal cycling/bakeout methodology and integration lessons. [10, 65] *Flight-model thermal cycling used two cycles from -15 to +50 degrees C. Bakeout reached 60 degrees C.*	Laminate/design discussion, passive isolation description, TVAC results and reported flight temperatures. [12, 13, 67] *V-R3x reported internal temperatures of 18.5 to 22.4 degrees C.*	ESATAN model, thermal geometry, materials/surfaces/interfaces and hot/cold analyses. [49, 54] *Supplier-driven redesign changed the PCB stack and added cover PCBs. [20]*	Battery-box construction/CAD, Kapton/Teflon/material descriptions and actual temperature datasets. [56, 68] *The reported BIRDS-3 temperature dataset was collected with the heater off.*
	Battery heater/sensor schematics, ChibiOS heater state machine and current thermal-spin/reorientation control source. [4, 5, 61] *Thermal reorientation uses the same TODO reaction-wheel command path.*	Sensor/hysteresis firmware and a separate incomplete heater-board release. [43, 51] *The separate HeaterBoard schematic contains one connector. Its PCB contains additional footprints. [64]*	EPS schematic PDF with heater/RTD circuits, PID/PWM/sensor firmware and heater bench tests. [66] *Heater firmware is enabled with a 5 degrees C setpoint.*	Current sensor schematic/BOM and temperature driver code. [14, 15] *The v05 temperature driver reads the BMX160 IMU.*	OBC/ADCS temperature interfaces and supporting software. [19, 21]	FAB heater/temperature schematic, BOM and threshold-control code. [69]
	New battery hardware/Zephyr application configuration and engineering notes. [62] *Published Zephyr defaults disable heaters.*					Later FAB firmware with the heater loop/handlers commented out. [25] *BIRDS-5 firmware comments out automatic heater control for power saving.*
Propulsion	No thruster/feed CAD, schematics, BOM or propulsion control code identified in scope. [2]	No propulsion CAD, thruster/feed schematic, BOM or control source identified in scope. [43]	No propulsion CAD, thruster/feed schematic, BOM or control source identified for FloripaSat-I. [10, 45]	No propulsion CAD, feed/thruster electronics, BOM or control source identified. [13]	No propulsion CAD, feed/thruster electronics, BOM or control source identified. The AOCS manual plans no maneuvers. [49]	No propulsion CAD, feed/thruster schematic, BOM or control code identified in the scoped platform release. [24]
Power	Solar v5.3 and battery v3 EAGLE schematics/PCBs/XLSX BOMs, MPPT/protection/switching firmware and assembly references. [2, 61, 70] *Fuel-gauge learning cycles were skipped on OreSat0 and OreSat0.5. The December 2024 update reports slowly discharging OreSat0.5 batteries. [40, 62]*	Altium EPS v6 schematic/PCB and PDF, MPPT/regulation/switching/protection firmware, solar/battery models and intended flight releases. [42, 51, 63] *The EPS schematic names STM32L152. Its README names STM32L100RCT6.*	EPS schematic PDF, MPPT/energy/protection firmware and battery/purchased-panel descriptions. [10, 65, 66] *The EPS wiki specifies five GomSpace P110 panels and one ISIS top panel.*	Platform power-design publication, historical circuitry and flight battery/energy results. [11–13] *V-R3x harvested roughly one tenth of expected solar power and remained power negative.*	2019 KiCad EPS schematic/PCB and BOM.xlsx. [20, 49, 73] *The CDR architecture uses commercial EPS/battery modules. The 2019 prototype retains design TODOs.*	OBC/EPS/FAB PDF schematics/BOM/firmware, native solar/backplane/access-board source and actual EPS datasets. [22, 68, 69, 74] *The BIRDS-4 report records 458 days of Tsuru operation and power-related loss of Maya-2 and GuaraniSat-1 within one week. [23]*
	Battery v4.1 and new solar/MCU KiCad designs, Zephyr applications and updated engineering notes. [62, 71] *Power releases now include battery v4.1, Zephyr applications and MCXN947 solar variants.*	Bench MPPT/heater experiments, Simulink/Simscape and LabVIEW test/model artifacts. [72]		Power.sch, protected battery-board schematic/PCB/BOM, charger/MPPT/monitoring drivers and manufacturing files. [14, 15, 52] *Flight setup calls for removing USB charging jumpers and selecting solar diode D2.* Beep-Sat battery task and low-power/shutdown example source. [17, 35] *The shutdown example leaves wakeup behavior as an exercise.*	Power-budget scripts/STK models and documented purchased EPS, battery and array architecture. [20, 49]	Separate firmware, parts lists and test/procedure package. [25, 26, 75] *BIRDS-5 OBC/EPS and FAB BOMs retain BIRDS-3 revision names.*

Table 2: Ground segment materials. Contribution groups are colored by status: Flight-linked , Changed or superseded , Unvalidated , Ground reference , Not found .

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Planning and scheduling	UniClOGS pass predictor/commander, pass-calculator code, tests and the published station architecture. [76, 77] *Inter-station coordination is manual.*	SatNOGS observation-job fetching and pass-scheduling source plus architecture documentation. [43, 80] Archived JavaScript commands for onboard scheduling and science-script control. [80] *The UPSat web extension was removed upstream in May 2017.*	Time-window/periodicity uplink_event.cpp, timed dispatch and Gpredict launch integration. [81]	Context only: tracking/operations publications and onboard tasko examples. [13, 35]	Yamcs ST[11] command definitions, release-time form and documented SatNOGS pass scheduling. [82–84] *The monitoring repository contains one sample DAG with schedule=None.*	Context only: tracking functions and onboard reservation firmware. [22, 85]
Spacecraft commanding	Yamcs shell/XTCE/YAML, EDL postprocessor, pass commander transport and matching C3 command handlers. [76, 86, 87]	ECSS command builder, browser UI, packet/large-data handlers, serial/UDP transport and historical transmit integration. [43, 80, 88] *The UPSat-specific SatNOGS client extension was removed upstream in May 2017.*	Qt NGHam command generation/dispatch, GNU Radio GFSK transmit flowgraph, Makefile and matching spacecraft handlers. [9, 10, 81] Published serial/server output code paths and GUI hooks. [81] *Selecting serial or server uplink raises an Under development error.*	Mission paper describing successful ground command/control and station architecture. [13] Ground-radio command/ACK examples and spacecraft cdh.py handlers. [17, 35] *Command examples use fixed passcodes and remote eval/exec.*	Yamcs postprocessor/XTCE/history, Angular validation/submission and serial/TCP bridges. [82, 83]	C# command UI, Transmit_async, ADDNICS/KISS/AX.25 framing and TNC serial/frequency control. [24, 85]
Ground equipment control	Stationd GPIO/PA/LNA/relay control, rotator/Doppler/safing source and tests, UPD KiCad/wiring and relay-driver BOM. [77, 89, 90] *The 2024 paper documents operating UniClOGS installations.*	Rotctld/rigctld observer workers, tracking/Doppler control and station setup documentation. [80]	SDR device/frequency/rate controls, GNU Radio configuration and Gqrx/Gpredict launch integration. [81]	V-R3x telescope-mount/antenna and tracking setup publication. [13] radio_helpers.py device configuration and receiver sleep/MQTT control code. [91] *Remote MQTT commands invoke eval/exec.*	Specialized SatNOGS rotator/EasyComm/homing/end-stop/motor firmware and integrated upstream hardware references. [49, 92]	Prosistel/Yaesu, radio mode/frequency/Doppler and TLE tracking source with textbook setup. [24, 85]
Space/ground communications	GNU Radio TX/RX flowgraphs, SDR/station code, interface electronics and RF wiring/configuration documentation. [77, 90, 93] *The RF chain uses commercial power amplifiers and SDR equipment.* Handheld receiver mechanical/electronic source, firmware and assembly material. [27]	USRP TX/RX flowgraphs, AX.25/QB50 framing/deframing and UDP transport source/configuration. [80, 88] *The transceiver script uses Python 2.*	GFSK TX/RX flowgraphs, WAV/UDP reception, sample recordings, support list and reception instructions. [32, 81] GRS NGHam framing/Reed-Solomon source. [81] *Reed-Solomon initialization sets nroots, mm and nn to zero.*	Published helical antenna, LNA, amplifier and bidirectional mission link description. [13] PyGS shopping list, wiring, assembly, spacer STL and receive code, plus Beep-Sat transmit examples. [17, 91] *PyGS publishes third-party library licenses only.*	CDR RF chain/LimeSDR/antenna design, uplink GMSK GNU Radio flowgraph and Yamcs frame interfaces. [49, 82, 94] *The UHF-downlink and S-band flowgraph directories contain README files only.*	KISS/AX.25/TNC serial software, Icom/KPC9612+/Yagi configuration and documented SatNOGS reception. [24, 26, 85]
Mission data processing	Handheld FEC/block reconstruction C code, receive scripts, MP4 handling and ffprobe stream checks. [27, 95] *Receiver documentation reports double-free and invalid-pointer crashes, handled by service restarts.*	Historical ECSS raw science-unit extraction, large-data reassembly, filenames and UTC handling. [80] *The science-log parser belongs to the removed UPSat-specific client extension.*	telemetry_data.cpp payload bytes and experiment-specific payload_x_status.cpp source. [81]	Dynamics/measurement formulation and Levenberg-Marquardt pseudocode in the V-R3x paper. [13] *Complete ranging and radiation datasets remained onboard after operations ended.*	Bench UI raw-camera image assembly, instrument plots and tests. [96] Yamcs TMParser.java segment reconstruction and custom PUS preprocessing. [82]	Context only: published mission images and raw-hex capture/logging. [23, 85]

Subsystem	OreSat	UPSat	FloripaSat-I	PyCubed	AcubeSAT	BIRDS
Mission data dissemination	Published telemetry web endpoint and mission ground data-access architecture. [77, 97] OreSat Live web/video server/player and PAWS frontend. [78, 95]	SatNOGS recording/waterfall upload source and public observation/telemetry references. [43, 80]	Public audio/data/image archive, SatNOGS observation references and GRS packet/CSV export code. [81, 98]	Context only: publications, raw MQTT relay and beacon-decoder links. [13, 91]	Angular bucket/file/image dialogs, list/upload/download/delete and Yamcs object/archive APIs. [82, 83]	Context only: published images, technical papers and actual EPS datasets. [23, 68, 85]
Network	Ethernet/VLAN/VPN architecture, station service/image setup, systemd network files and TCP/UDP interfaces. [77, 87, 99] *The 2024 paper describes deployed station networking.*	HTTP job/upload API, TCP Hamlib and UDP radio source, with architecture/install documentation. [80] Redis/Socket.IO transport belonging to the historical UPSat web interface. [80] *The web-specific transport was removed from upstream SatNOGS.*	udp_server.cpp and udp_decoder.cpp source. [81] Published external-server uplink hooks. [81] *The external-server uplink path raises an Under development error.*	WiFi/socket/MQTT publish-subscribe source, offline cache and later upload logic. [91] *The example uses unencrypted MQTT and an external broker.*	Yamcs TCP/HTTP/WebSocket connections, mission YAML, Docker instructions and serial/TCP bridges. [82, 83]	BIRDS ground-network/SatNOGS integration descriptions. The released C# application mainly uses local files and serial devices. [26, 85]
Telemetry processing	Beacon schemas/decoders, SatNOGS import, Yamcs preprocessing/tests, XTCE databases, archive and display configuration. [86, 87, 100]	ECSS/WOD decoding and engineering conversion in the historical UPSat client extension. [43, 80] *The UPSat decoder was removed from upstream SatNOGS.* Standalone WOD plotting source and example JSON data. [101] *Example datasets are ground-test data dated 2016.*	Packet/engineering conversion, display/logging, CSV plotting and sample recordings. [10, 81] GRS FEC implementation and standalone Python decoder helpers/GUI. [81, 102] *The standalone decode callback only loads a WAV file and logs its sample rate.*	Published V-R3x beacon layout and reception/telemetry results. [13, 67] CRC/hex/JSON handling with time/station/RSSI metadata. [91]	XTCE databases, preprocessing, recorder/alarm/replay configuration and live/archive graphs. [82, 83] New archiving-service template and monitoring example DAG. [84, 103]	Actual EPS datasets and their calibration/processing publication. [68] C# serial/KISS intake, hex conversion, packet counts and log export. [85]

Table 3: Sources cited in the materials tables. Repository URLs identify the reviewed reference or branch where available.

Ref.	Source
[1]	OreSat Project. oresat-c3-hardware @ v5_0, . URL https://github.com/oresat/oresat-c3-hardware/tree/v5_0. Accessed 3 November 2026
[2]	Portland State Aerospace Society. OreSat0: mission history and subsystem manifest, . URL https://www.oresat.org/satellites/oresat0. Accessed 3 November 2026
[3]	OreSat Project. oresat-c3-hardware @ 738a430, . URL https://github.com/oresat/oresat-c3-hardware/tree/738a430a2270f489b3ff0aad043578f60fcf075e. Accessed 3 November 2026
[4]	OreSat Project. oresat-c3-software @ 1136e31, . URL https://github.com/oresat/oresat-c3-software/tree/1136e319fb85844638c0c549663045f731468e7f. Accessed 3 November 2026
[5]	OreSat Project. oresat-firmware @ c3_v1_1, . URL https://github.com/oresat/oresat-firmware/tree/c3_v1_1/src/f4/app_control/source. Accessed 3 November 2026
[6]	Libre Space Foundation. upsat-obc-hardware @ 4661719, . URL https://gitlab.com/librespacefoundation/upsat/upsat-obc-hardware/-/tree/4661719549217c3c32389e0c80aeca8c0dc1eb12. Accessed 3 November 2026
[7]	Libre Space Foundation. upsat-obc-software @ a04902f, . URL https://gitlab.com/librespacefoundation/upsat/upsat-obc-software/-/tree/a04902fc894e9c22d39a4b5e92af9874f256e941. Accessed 3 November 2026
[8]	Libre Space Foundation. upsat-ecss-services @ 616003e, . URL https://gitlab.com/librespacefoundation/upsat/upsat-ecss-services/-/tree/616003e52e2ab282054b89c1dfac0b4bd0f96e1b. Accessed 3 November 2026
[9]	FloripaSat Team. obdh @ e24bf3e, . URL https://github.com/floripasat/obdh/tree/e24bf3e6395eafdae88358c90d23c096502167ff. Accessed 3 November 2026
[10]	Qualification and validation test methodology of the open-source CubeSat FloripaSat-I, 2020. URL https://www.jseepub.com/article/2020/1004-4132/20200615.shtml. Accessed 3 November 2026
[11]	PyCubed Project. hardware @ d1adfd0, . URL https://github.com/pycubed/hardware/tree/d1adfd0c207dd5951ab175af53b3b7c1c72a5212/mainboard-v01. Accessed 3 November 2026
[12]	PyCubed Project. PyCubed: An Open-Source, Radiation-Tested CubeSat Platform Programmable Entirely in Python, 2019. URL https://github.com/pycubed/documentation/blob/0670db47f1a26204e200c8a92beb2f431a1d65cb/PyCubed_smallsat-paper.pdf. Accessed 3 November 2026
[13]	The V-R3x Mission: Towards Autonomous Networking and Navigation for CubeSat Swarms, 2022. URL https://roboticexplorationlab.org/papers/vr3x.pdf. Accessed 3 November 2026
[14]	PyCubed Project. hardware @ d1adfd0, . URL https://github.com/pycubed/hardware/tree/d1adfd0c207dd5951ab175af53b3b7c1c72a5212/mainboard-v05. Accessed 3 November 2026
[15]	PyCubed Project. software @ 72ab3a1, . URL https://github.com/pycubed/software/tree/72ab3a1e1869b4d8a6096102e156567f89b0059b/mainboard-v05. Accessed 3 November 2026
[16]	PyCubed Project. NOTICE: August 1 2024, SAMD51 proton sensitivity, 2024. URL https://github.com/pycubed/forums/discussions/87. Accessed 3 November 2026
[17]	PyCubed Project. software_example_beepsat @ c057fa0, . URL https://github.com/pycubed/software_example_beepsat/tree/c057fa050b28df7fc0a898416409a4c1dc8d325d. Accessed 3 November 2026
[18]	AcubeSAT Team. obc-pcb @ adcs-obc-v2.5, . URL https://gitlab.com/acubesat/obc/obc-pcb/-/tree/adcs-obc-v2.5. Accessed 3 November 2026
[19]	AcubeSAT Team. embedded-software @ main, . URL https://gitlab.com/acubesat/obc/embedded-software/-/tree/main. Accessed 3 November 2026
[20]	Lessons learned from the design and early manufacturing phases of AcubeSAT, 2026. URL https://mediatum.ub.tum.de/doc/1857487/1857487.pdf. Accessed 3 November 2026
[21]	AcubeSAT Team. obc-pcb @ v3.1, . URL https://gitlab.com/acubesat/obc/obc-pcb/-/tree/v3.1. Accessed 3 November 2026
[22]	BIRDS Open Source Project. BIRDS4-OBC @ 490f744, . URL https://github.com/BIRDSOpenSource/BIRDS4-OBC/tree/490f74487d65ebfb0037bbdd9f3e7aab6aefc3a5. Accessed 3 November 2026
[23]	BIRDS-4 1U CubeSat Missions and On-Orbit Results. URL https://nanosat11th.itu.edu.tr/assets/papers/BIRDS-4.pdf. Accessed 3 November 2026
[24]	Kyushu Institute of Technology. BIRDS Platform Interface Control Document, English Rev F. URL https://github.com/BIRDSOpenSource/BIRDS-GeneralDocumentation/blob/main/BIRDS_Platform_ICD_ENGLISH_RevF_0303.pdf. Accessed 3 November 2026
[25]	BIRDS Open Source Project. BIRDS5-Source_Codes @ 8dc0580, . URL https://github.com/BIRDSOpenSource/BIRDS5-Source_Codes/tree/8dc0580ba154e26f59eb1dfcd5d3bcc1e233c8ee. Accessed 3 November 2026
[26]	BIRDS Project. Lessons Learnt and Findings from BIRDS5-FTA, 2023. URL https://birds-project.com/open-source/pdf/20230308_Birds5_project_OpenSourceWebinar.pdf. Accessed 3 November 2026
[27]	OreSat Project. oresat-live-handheld-ground-station, . URL https://github.com/oresat/oresat-live-handheld-ground-station. Accessed 3 November 2026
[28]	Libre Space Foundation. upsat-comms-hardware @ v3, . URL https://gitlab.com/librespacefoundation/upsat/upsat-comms-hardware/-/tree/v3. Accessed 3 November 2026
[29]	Libre Space Foundation. upsat-comms-hardware @ v3.1, . URL https://gitlab.com/librespacefoundation/upsat/upsat-comms-hardware/-/tree/v3.1. Accessed 3 November 2026
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